


유형 5 세항식의 합성

○ 021) $f(x) = 2x - 1, g(x) = -x + 1, h(x) = -2x^2$

① $h \circ (g \circ f)$

$$g \circ f = -(2x - 1) + 1 = -2x + 2$$

$$h \circ g \circ f = -2(-2x + 2)^2$$

② $(h \circ g) \circ f$

$$h \circ g = -2(-x + 1)^2$$

$$h \circ g \circ f = -2(-(2x - 1) + 1)^2 = -2(-2x + 2)^2 = -4(x - 1)^2$$

동등함

○ 022) $f(x) = x + 1, g(x) = 2x, h(x) = x^2$

$h \circ g \circ f(x)$

$$g \circ f = 2(x + 1) = 2x + 2$$

$$h \circ g \circ f = (2x + 2)^2$$

○ 023) $f(x) = x - 2, g(x) = 3x, h(x) = x + 5$

$h \circ g \circ f(x)$

$$h \circ g = 3x + 5$$

$$h \circ g \circ f = (3x + 5) - 2 = 3x + 3$$

○ 024) $f(x) = x^2, g(x) = x + 1, h(x) = 2x$

$f \circ g \circ h(x)$

$$g \circ h = 2x + 1$$

$$f \circ g \circ h(x) = (2x + 1)^2$$

$$f \circ g \circ h(1) = 3^2 = 9$$

✗ 025) $f(x) = x + a, g(x) = 2x, h(x) = x^2$

$(h \circ g \circ f)(0) = 16$ 일 때 a 의 값?

$$h \circ g = (2x)^2 = 4x^2$$

$$h \circ g \circ f(x) = 4x^2 + a$$

$$h \circ g \circ f(0) = 4 \cdot 0^2 + a = 16, a = 16$$

유형 6 합성함수의 미정계수

○ 026) $f(x) = 2x + 1, g \circ f(x) = 6x + 5$

$$g(x) = 3(2x + 1) + 2 = 3x + 2$$

○ 027) $f(x) = x - 3, (f \circ g)(x) = 2x + 1$

$$(\quad) - 3 = 2x + 1$$

$$g(x) = 2x + 4$$

○ 028) $g(x) = 3x - 1, (g \circ f)(x) = 6x + 2$

$$\rightarrow 3(\quad) - 1 = 6x + 2$$

$$= 3(2x + 1) - 1 = 6x + 2$$

$$f(x) = 2x + 1$$

○ 029) $f(x) = ax + b, f \circ f(x) = 4x + 9$

$$= 2x + 3$$

$$= 2(2x + 3) + 3$$

$$a = 2, b = 3$$

○ 030) $f(x) = 2x + 3, (f \circ g)(x) = 4x - 1$

$$g(2)$$

$$\rightarrow 2(2x - 2) + 3 = 4x - 1$$

$$g(x) = 2x - 2$$

$$g(2) = 4 - 2 = 2$$

유형 7) 항등함수 ($y=x$)

○ 031) $f(x) = 2ax - 2$ 가 $f \circ f = I$ 일때 a ?

$$= 2a(2ax - 2) - 2 = x$$

$$= 4a^2x - 4a - 2 = x \rightarrow x=0 \text{ 일때도 성립}$$

$$= -4a - 2 = 0$$

$$= a = -\frac{1}{2}$$

✗ 032) $f(x) = ax + b$ $f \circ f = I$ 일때 a, b 의 조건

$$= a(ax + b) + b = x$$

$$= a^2x + ab + b = x$$

○ 033) $f(x) = 3x + 2$ $(g \circ f)(x) = x$

$$a(3x + 2) + b = x$$

$$\frac{3ax + 2a + b}{1} = x$$

$$a = \frac{1}{3} \quad b = -\frac{2}{3}$$

$$g(x) = \frac{1}{3}x - \frac{2}{3} = \frac{x-2}{3}$$

○ 034) $f(x) = \frac{x}{x-1} \quad (x \neq 1)$

① $f \circ f = \frac{\frac{x}{x-1}}{\left(\frac{x}{x-1}\right) - 1} = \frac{\frac{x}{x-1}}{\frac{x - (x-1)}{x-1}} = \frac{\frac{x}{x-1}}{\frac{1}{x-1}} = \frac{x}{x-1} = x$

② $f \circ f = I$ 이다

○ 035) $f \circ g = g \circ f$ 일지 판별 및 증명

$f(x) = 2x + 1, g(x) = x + 2$ 라고 하면

① $f \circ g = 2(x + 2) + 1 = 2x + 5$

② $g \circ f = (2x + 1) + 2 = 2x + 3$

① $\neq$ ② 이므로 $f \circ g = g \circ f$ 는 항상 성립하지 않는다.
→ 교환 법칙이 적용되지 않는다

유형 8) 합성함수 항등함수

✗ 036) $f(x) = x^2 - 2x$
 $(f \circ g)(x) = x^2 - 1$ 일때 1차함수 $g(x)$?

○ 037) $f(x) = 2x + 1$ 일때 $f^n(x)$?

$$f' = 2x + 1$$

$$f^2 = 2(2x + 1) + 1 = 4x + 3$$

$$f^3 = 2(4x + 3) + 1 = 8x + 7$$

$$f^n = 2^n x + (2^n - 1)$$

✗ 038) $f(x+1) = 2f(x) + 1$
 $f(0) = 0$ 일때 $f(3)$ 은?

○ 039) $f(x) = ax + b$ 가 $f^3(x) = 8x + 7$
 a, b 의 값? ($a > 0$)

$$f' = ax + b$$

$$f^2 = a(ax + b) + b = a^2x + ab + b$$

$$f^3 = a(a^2x + ab + b) + b$$

$$= \frac{a^3x + a^2b + ab + b}{8} = \frac{a^3x + a^2b + ab + b}{7}$$

$$a = 2 \leftarrow \frac{a^3}{8} \quad \therefore a = 2, b = 1$$

$$4b + 2b + b = 7 \quad b = 1$$

○ 040) $f(x) = x + 2, g(x) = 3x$
 $(g \circ f)(x) = 12$ 일때 x 는?

$$3(x + 2) = 12$$

$$3x + 6 = 12$$

$$x = 2$$